Module 7

Capital Budgeting

FINE 3010, Financial Management

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- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project
- **Capital budgeting**, the process of analyzing capital investment opportunities ("projects") and deciding which ones to accept

Module 7, Part 1

Project Cash Flow

Recall from Module 2, the Statement of Cash Flows

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- **Operating activities**, the same idea, project net income plus depreciation, with the change in NWC as its own component
- **Investing activities**, the project's capital investments, buying its assets at the start and selling them at the end
- **Financing activities**, none, we evaluate the project as if it were all-equity financed, so no interest

Identify Project Cash Flow

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$$
\begin{aligned}
\text{Cash flow from a project} &= \text{Project net income} \\
&+ \text{Depreciation} \\
&+ \text{Cash flow from capital investments} \\
&+ \text{Cash flow from changes in working capital}
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Project EBIT
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- Note, revenue minus cash expenses is called the "gross profit"

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- So add it back, as in **Module 2**'s operating activities,

$$\text{Adjusted accounting profits} = \text{Project net income} + \text{Depreciation}$$

Example, Project Net Income and Depreciation

- A project requires an investment of $2,000, producing gross profits (revenues less cash expenses) of $1,100 in year 1 (and $1,300 in year 2). The initial investment can be depreciated at $1,000 per year. The corporate tax rate is 21%

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- Calculate the project net income plus depreciation for year 1

Example, Project Net Income and Depreciation

- Solution,

Item	$
Gross profit	1,100
Depreciation	1,000
Project EBIT	100
Taxes	21
Project net income	79
+ Depreciation	1,000
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- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$
- Project net income + Depreciation $= 79 + 1{,}000 = \$1{,}079$

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- If the project requires spending money on investment to get started (e.g., purchasing a building), that is a negative cash flow
- Usually capitalized and depreciated over time from an accounting perspective, in this course always "straight-line" depreciation,

$$\text{Annual depreciation} = \frac{\text{Investment} - \text{Salvage value}}{\text{Expected life}}$$

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- Working capital is often recovered at the end of the project (inventories consumed, customers pay their bills), a positive cash flow

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Module 7, Part 2

Putting It All Together

Example, Putting It All Together

- A company is considering building a new data center which costs \$1 billion (\$1,000 million) and will last 5 years. It will be depreciated straight-line over 5 years with a salvage value of \$200 million. The data center can also be sold at $t = 5$ for \$200 million

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- The corporate tax rate is 21%, and the discount rate is 8%
- Should the company build the data center or not?

Example, Putting It All Together

- Solution ($ millions),

Year	0	1	2	3	4	5
Gross profit		200	200	200	200	200
Depreciation		160	160	160	160	160
EBIT		40	40	40	40	40
Net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		160	160	160	160	160
Operating cash flows		191.6	191.6	191.6	191.6	191.6
Capital investment	−1,000					
Sale of the data center						200
Working capital	−100					100
Total cash flows	−1,100	191.6	191.6	191.6	191.6	491.6
NPV	−130.82					

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  - **Operating cash flow before accounting for ΔNWC**, in years 1–5 the rental income adds 200 to gross profits. Depreciation rises by $(1{,}000 - 200)/5 = 160$ per year, a net 40 higher EBIT. After taxes and adding back depreciation, operating cash flow rises by 191.6 each year

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- Putting it all together, NPV $= -\$130.82$ million, so the company should not build the data center

Module 7, Part 3

The Principles behind Project Cash Flows

General Principles for Identifying Project Cash Flow

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3. Don't forget about depreciation and taxes
4. Only consider incremental cash flows

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- In the data center table, we discounted the operating cash flows ($191.6 million a year), not the net income ($31.6 million)

Cash Flow vs. Accounting Income

- A project costs $2,000 and is expected to last 2 years, producing cash income of $1,500 and $500, respectively. The cost can be depreciated at $1,000 per year. Given a 10% required return, compare the NPV using accounting income to the NPV using cash flow

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- This project is clearly a loser!

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  - The effects of interest (and the interest tax shield) enter when we *separately* consider financing decisions
- The data center table has no interest line, even if the company borrows to build it

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- Different depreciation schedules, in this course we stick to "straight-line" depreciation, as in the data center, $(1{,}000 - 200)/5 = \$160$ million a year

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$$\text{Incremental cash flow} = \text{Cash flow with the project} - \text{Cash flow without it}$$

- In the data center table, the rental income and the spare parts exist only because of the data center, so they count

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f. Remember terminal cash flows

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Rule a, Indirect Effects

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- But the launch will hurt sales of the older console, forecast to create \$800m and \$1,000m in years 1 and 2, only \$500m and \$400m with the new one on the market
- What is the **incremental** cash flow of the new console?





Rule a, Indirect Effects

- Solution ($ millions),

	$t = 0$	$t = 1$	$t = 2$
Cash flow from the old console without the new one		800	1,000
Cash flow from the old console with the new one		500	400
Cash flow from investment in and sales of the new one	-200	400	800
Incremental cash flow of the new console	-200	$400 - 300 = 100$	$800 - 600 = 200$

Rule b, Sunk Costs

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- They do not affect incremental cash flows, so they do not affect the NPV of a decision. Ignore them!
- Many sunk costs are "related" to a project, but no longer incremental if the current decision does not affect them



Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
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- Is this project a good idea today?
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- E.g., you plan a new factory building that costs $10m to construct, on a piece of land you already own (and do not need for anything else). Alternatively, you could sell the land for $3m
- What is the actual cost of building the factory?

Rule d, Overhead

- **Overhead**, ongoing administrative expenses of a business that cannot be attributed to any specific business activity, but are still necessary for the business to function
  - E.g., rent and utilities of the central office, top management salaries

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- The principle of incremental cash flows says only *changes* in overhead that result from a project count

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  - Only changes in NWC that are incremental to the project count
- Working capital often keeps changing during the project, and is often recovered at the end
- Common mistakes,
  1. Forgetting about working capital entirely
  2. Forgetting that working capital may change during the life of the project
  3. Forgetting that working capital is often recovered at the end of the project

Rule f, Terminal Cash Flows

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  - E.g., selling the machines and equipment of the project, a positive cash flow!

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$$(500 - 200) \times 21\% = \$63 \text{ million}$$

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$$(500 - 200) \times 21\% = \$63 \text{ million}$$

- The net (after-tax) proceeds are $500 - 63 = \$437$ million, and now the NPV is positive ($+\$30.48$ million)

Example, the Data Center with a Higher Terminal Cash Flow

- Solution ($ millions),

Year	0	1	2	3	4	5
Gross profit		200	200	200	200	200
Depreciation		160	160	160	160	160
EBIT		40	40	40	40	40
Net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		160	160	160	160	160
Operating cash flows		191.6	191.6	191.6	191.6	191.6
Capital investment	−1,000					
Sale of the data center (after tax)						437
Working capital	−100					100
Total cash flows	−1,100	191.6	191.6	191.6	191.6	728.6
NPV	30.48					
IRR	8.8%					

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- E.g., oil wells require "capping" when they no longer produce. Capping is expensive, so oil well projects are worth more if you do not cap them (although this is illegal!)
- But apparently many companies still do this, Bloomberg, "It Pays to Not Pay Your Debts"



Practice Problem

- A firm plans a new manufacturing plant on a site it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. The market value of the site
b. The market value of the existing buildings
c. Demolition costs and site clearance
d. The cost of a new access road put in last year
e. Lost cash flows on other projects due to executive time spent on the new facility
f. Future depreciation of the new plant
g. The reduction in the firm's tax bill resulting from depreciation of the new plant
h. The initial investment in inventories of raw materials
i. Money already spent on engineering design of the new plant

Practice Problem

a, b. The site and buildings could have been sold or put to another use. Their values are opportunity costs, treated as incremental cash outflows

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i. The expenditure on design is a sunk cost and not an incremental cash flow

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- Count indirect effects, opportunity costs, working capital, and terminal cash flows (with the tax on any gain over book value), ignore sunk costs and allocated overhead

Module 7, Part 4

Full-Scale Capital Budgeting Problems

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - It costs 1,000 and lasts 5 years, depreciated straight-line to a salvage value of 200, and sold at $t = 5$ for 200

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- This time, build the table step by step

Example, the Data Center Again

- Solution ($ millions), the operating cash flows first, then working capital, then the capital investment and the sale, then the totals

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Operating cash flows		191.6	191.6	191.6	191.6	191.6
Working capital	−100					100
Capital investment	−1,000					
Sale of the data center						200
Total cash flows	−1,100	191.6	191.6	191.6	191.6	491.6
NPV	−130.82					

- NPV = −$130.82 million, reject the project, the same answer as in Part 2

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3. Add the capital investment at the start and the terminal cash flows at the end (after tax)
4. Total the cash flows year by year, discount at the discount rate, and apply the NPV rule

Example 1, Growing Sales and a Taxed Sale of Equipment

- A company makes bottles and just developed a new bottle design
- The initial investment in machinery is $6 million. The equipment is depreciated straight-line over 6 years to zero, but will actually be sold in 6 years for $620,000 (so there are tax consequences)

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- The new design is phased out after 6 years. The tax rate is 21%, and the required rate of return is 12%
- What is the project's NPV?

Example 1, Growing Sales and a Taxed Sale of Equipment

- Operating cash flows ($ thousands),

Year	1	2	3	4	5	6
Sales (thousand bottles)	1,000	1,050	1,103	1,158	1,216	1,276
Sales ($)	4,000	4,200	4,410	4,631	4,862	5,105
Cost ($)	2,000	2,100	2,205	2,315	2,431	2,553
Depreciation	1,000	1,000	1,000	1,000	1,000	1,000
EBIT	1,000	1,100	1,205	1,315	1,431	1,553
Taxes	210	231	253	276	301	326
Net income	790	869	952	1,039	1,130	1,227
+ Depreciation	1,000	1,000	1,000	1,000	1,000	1,000
Operating cash flows	1,790	1,869	1,952	2,039	2,130	2,227

Example 1, Growing Sales and a Taxed Sale of Equipment

- Terminal cash flow from the sale of the equipment ($ thousands), the book value is zero so the whole sale price is taxed,

Terminal cash flows	$
Sales price	620
Tax	130
Net income from sale	490

Example 1, Growing Sales and a Taxed Sale of Equipment

- Total cash flows ($ thousands),

Year	0	1	2	3	4	5	6
Capital	−6,000						490
Working capital	−250						250
Operating cash flows		1,790	1,869	1,952	2,039	2,130	2,227
Total cash flows	−6,250	1,790	1,869	1,952	2,039	2,130	2,966
PV	−6,250	1,598	1,490	1,389	1,296	1,209	1,503
NPV	2,235						

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PV	−6,250	1,598	1,490	1,389	1,296	1,209	1,503
NPV	2,235						

- The Excel solution is in the "Bottles" tab of Module 7 Excel Example.xlsx

Practice Problem

- A firm has developed a new mousetrap. It can go into production for an initial investment in equipment of \$6.3 million. The equipment will be depreciated straight-line over 6 years, but in fact can be sold after 6 years for \$695,000
- Working capital at each date must be maintained at 10% of next year's forecast sales. Production costs are \$1.80 per trap and the traps sell for \$7 each. The project ends in 6 years when the trap becomes obsolete. The tax rate is 40%, and the required rate of return is 11%

Year	0	1	2	3	4	5	6	After
Sales (million traps)	0	0.6	0.8	0.9	0.9	0.5	0.2	0

Practice Problem

- Suppose the firm can cut its working capital requirements in half with better inventory control. By how much does this increase the project NPV?

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- Solution ($ millions), working capital at each date is 10% of next year's sales,

Year	0	1	2	3	4	5	6
Sales units		0.600	0.800	0.900	0.900	0.500	0.200
Revenue	0.000	4.200	5.600	6.300	6.300	3.500	1.400
NWC	0.420	0.560	0.630	0.630	0.350	0.140	0.000
Cash flow from NWC	−0.420	−0.140	−0.070	0.000	0.280	0.210	0.140

Practice Problem

- With working capital requirements halved ($ millions),

Year	0	1	2	3	4	5	6
New NWC	0.210	0.280	0.315	0.315	0.175	0.070	0.000
New cash flow from NWC	−0.210	−0.070	−0.035	0.000	0.140	0.105	0.070
Change in cash flow	0.210	0.070	0.035	0.000	−0.140	−0.105	−0.070

Practice Problem

- With working capital requirements halved ($ millions),

Year	0	1	2	3	4	5	6
New NWC	0.210	0.280	0.315	0.315	0.175	0.070	0.000
New cash flow from NWC	−0.210	−0.070	−0.035	0.000	0.140	0.105	0.070
Change in cash flow	0.210	0.070	0.035	0.000	−0.140	−0.105	−0.070

- Discounting the changes in cash flow at 11%, the change in NPV is $0.1095 million

Practice Problem

- A firm may upgrade its modem pool. It last upgraded 2 years ago, when it spent $140 million on equipment with an assumed life of 5 years and an assumed salvage value of $20 million for tax purposes. The firm uses straight-line depreciation. The old equipment can be sold today for $100 million
- A new modem pool can be set up today for $210 million. It has a 3-year life and is depreciated to zero straight-line. The new equipment increases sales by $28 million per year and decreases operating costs by $14 million per year. At the end of 3 years it is worthless. The tax rate is 30% and the discount rate is 9%

Practice Problem

a. What is the net cash flow at time 0 if the old equipment is replaced?
b. What are the incremental cash flows in years 1, 2, and 3?
c. What is the NPV of the replacement project?
d. What is the IRR of the replacement project?

Practice Problem

- a. Annual depreciation is $(\$140 - \$20)/5 = \$24$. Book value at the time of sale is $\$140 - 2 \times \$24 = \$92$

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- a. Annual depreciation is $(\$140 - \$20)/5 = \$24$. Book value at the time of sale is $\$140 - 2 \times \$24 = \$92$

Sales price = $100, so net-of-tax proceeds from the sale are:
$100 − [0.30 × ($100 − $92)] = $97.60

Therefore, the net cash outlay at time 0 is $210 − $97.60 = −$112.40.

b. The project saves $14 in operating costs and increases sales by $28. The depreciation expense for the new machine would be $70 per year.

Therefore, including the depreciation tax shield, operating cash flow increases by:
($28 + $14) × (1 − 0.30) + ($70 × 0.30) = $50.40 per year

c. NPV = −$112.40 + $50.40 × ((1 ÷ 0.09) − {1 ÷ [0.09(1 + 0.09)^3]})
= $15.18

d. To find the internal rate of return, find the interest rate "r" that makes the Present Value of the inflows equal and opposite to the initial outflow. To do this set the PV of the three year annuity to $112.40 and solve for the discount rate (r):

$112.40 = $50.40 × [1 ÷ r − 1 ÷ r × (1 + r)^3]
r = 16.43%

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- Lay the years out in a table, total the cash flows, discount, and apply the NPV rule

Thank you!

Please let me know if you have any questions.